CHAPTER 7 - Sampling Distributions

SDS188

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1 Introduction

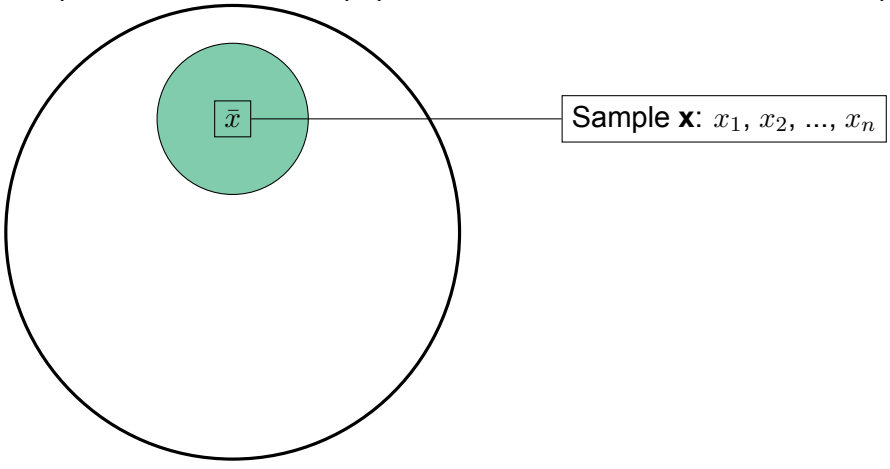
We use the sample statistic to estimate population parameters.

How do we estimate the unknown population parameters?

Calculate the sample statistic

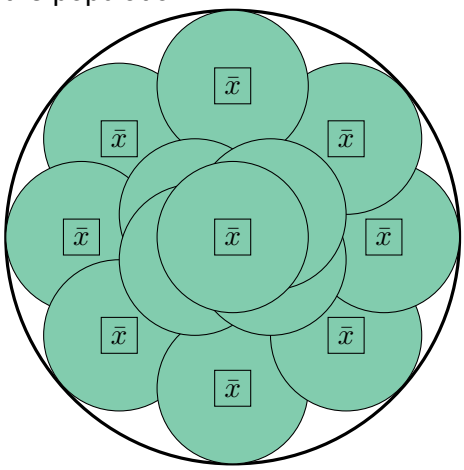
BUT

The statistic is calculated for that specific sample and does not give any information about the other samples selected from the population. A solution for this is the development of a sampling distribution.



For the sample, one can calculate the $\bar{x}$, but this only provides information about the one randomly

selected sample (with or without replacement). The same sample will result in the same mean, which is likely not close to the population parameter (Slide 6). What about the other possible samples from the population?



Each sample has its own sample mean.

X:

Sample 1 = x_1, x_2, ..., $x_n \longrightarrow \bar{x}$

Sample 2 = x_1, x_2, ..., $x_n \longrightarrow \bar{x}$

$\vdots$

Sample ... = x_1, x_2, ..., $x_n \longrightarrow \bar{x}$

The means of each sample can be seen as a data set, $\bar{x}_1$, $\bar{x}_2$, ...

The population or access to the population is not always available. How do we get the means of the other samples?

SOLUTION $\longrightarrow$ sampling distributions

Sampling distributions allow us to use one statistic to estimate the parameter. **The sampling distribution is the distribution of the calculated measures of all of the possible samples from the population**.

1.1 Standard error

As with any distribution, the sampling distribution is described by an expected value and a variance. We see that the different samples result in different means.

How do we measure the variation among the means?

The usual measure of variation that we would have used, is the standard deviation. For the sampling distribution, we use the standard error.

Suppose that the samples are selected from an infinite population with or without replacement, the standard error of the mean is calculated as:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

The standard error is influenced by the sample size (n)

$$n \uparrow, \sigma_{\bar{X}} \downarrow$$

$$n \downarrow, \sigma_{\bar{X}} \uparrow$$

THUS for the estimator $\bar{X}$, $E(\bar{X}) = \mu_{\bar{X}} = \mu$, $Var(\bar{X}) = \sigma^2_{\bar{X}} = \frac{\sigma^2}{n}$

2 Sampling distribution of the sample mean

Parameter:

$$\mu$$

2.1 Assumptions

- The population $X \sim N(\mu, \sigma^2)$
- σ is known

2.2 The distribution of the sample mean

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

The parameters of the normally distributed X can be used to develop a sampling distribution.

- $E(\bar{X}) = E(X) = \mu$
- $Var(\bar{X}) = \frac{\sigma^2}{n}$

*Here are some short proofs if this is ever asked:

$$\begin{aligned} E(\bar{X}) &= E(\frac{1}{n}\sum X) \\ &= \frac{1}{n}\sum E(X) \\ &= \frac{1}{n}\sum \mu \\ &= \frac{1}{n}(n\mu) \\ &= \frac{n\mu}{n} \\ &= \mu \end{aligned}$$

$$
\begin{aligned}
Var(\bar{X}) &= Var(\frac{1}{n}\sum X) \\
&= \frac{1}{n^2}\sum Var(X) \\
&= \frac{1}{n^2}\sum \sigma^2 \\
&= \frac{1}{n^2}(n\sigma^2) \\
&= \frac{\sigma^2}{n}
\end{aligned}
$$

2.3 Z-values

Since $\bar{X}$ is normally distributed, we can also use Z-values to calculate the sampling distribution of $\bar{X}$.

$$
Z = \frac{X-\mu}{\sigma} \longrightarrow Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}
$$

2.4 Probabilities

Typical questions for this chapter requires the calculation of probabilities.

Steps:

1. $P(\bar{X} < \bar{x})$ (Can draw the curve like in Ch 6)
2. Transform $\bar{x}$ into $Z \longrightarrow P(Z < z)$
3. Calculate the probability by means of the Z-table, HP calculator or Excel

HP calculator: $z \longrightarrow$ 

Excel: =NORM.S.DIST(z,TRUE) OR =NORM.DIST($\bar{x}$,μ,$\frac{\sigma}{\sqrt{n}}$,TRUE)

2.5 Interval that contains a percentage of sample means

Between which values do 95% of the sample means lie?

Given: %, μ, σ, n, and the assumption that $X \sim N(\mu, \sigma^2)$

Lower limit	$\bar{X}_L = \mu + (Z \times \frac{\sigma}{\sqrt{n}})$
Upper limit	$\bar{X}_U = \mu + (Z \times \frac{\sigma}{\sqrt{n}})$

The Z-values being referred to here depend on the percentage and whether it is the lower or upper limit. This can be determined by means of the Z-table, HP calculator or Excel.

HP calculator: $P(Z < z) \longrightarrow$ 

Excel: =NORM.S.INV($P(Z < z)$)

<u>If we are interested in 95% of the observations:</u>

For $\bar{X}_L$: The corresponding Z-value for $\frac{0.05}{2} = 0.025$

The Z-value for 0.025 is $z_{\frac{\alpha}{2}} = -1.96$

$$\bar{X}_L = \mu + (-1.96 \times \frac{\sigma}{\sqrt{n}})$$

For $\bar{X}_U$: The corresponding Z-value for $1 - \frac{0.05}{2} = 0.975$

The Z-value for 0.975 is $z_{1-\frac{\alpha}{2}} = 1.96$

$$\bar{X}_U = \mu + (1.96 \times \frac{\sigma}{\sqrt{n}})$$

95% of the sample means lie between $\bar{X}_L$ and $\bar{X}_U$

<u>If we are interested in 90% of the observations:</u>

For $\bar{X}_L$: The corresponding Z-value for $\frac{0.1}{2} = 0.05$

The Z-value for 0.05 is $z_{\frac{\alpha}{2}} = -1.645$

$$\bar{X}_L = \mu + (-1.645 \times \frac{\sigma}{\sqrt{n}})$$

For $\bar{X}_U$: The corresponding Z-value for $1 - \frac{0.1}{2} = 0.95$

The Z-value for 0.95 is $z_{1-\frac{\alpha}{2}} = 1.645$

$$\bar{X}_U = \mu + (1.645 \times \frac{\sigma}{\sqrt{n}})$$

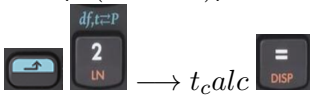
90% of the sample means lie between $\bar{X}_L$ and $\bar{X}_U$

2.6 What if sigma is unknown?

$$\bar{X} \sim t_{n-1}$$

$$T = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

Calculation of probabilities ($P(t < t_{calc})$):

HP calculator: df $\longrightarrow$  $\longrightarrow$ t_calc

Excel: =T.DIST(t_{calc},df,TRUE)

3 Sampling distribution van sample proportion

Parameter:

$$\pi$$

3.1 Assumptions

- The population $X \sim$ binomial(n, π)
- $n \times \pi > 5$ en $n \times (1 - \pi) > 5$

3.2 The distribution of the sample proportion

$$p \sim N(\pi, \frac{\pi \times (1 - \pi)}{n})$$

The parameters of the normally distributed p can be used to develop the sampling distribution.

- $E(p) = \pi$
- $Var(p) = \sqrt{\frac{\pi \times (1-\pi)}{n}}$

3.3 Z-values

Since p is normally distributed, one can also calculate Z-values for the sampling distribution.

$$Z = \frac{p - \pi}{\sqrt{\frac{\pi \times (1-\pi)}{n}}}$$

3.4 Probabilities

Typical questions for this chapter requires the calculation of probabilities.

Steps:

1. $P(p < p)$
2. Transform p into $Z \longrightarrow P(Z < z)$
3. Calculate probability by means of Z-table, HP calculator or Excel

HP calculator: $z \longrightarrow$ 

Excel: =NORM.S.DIST(z,TRUE) OR =NORM.DIST($\bar{x}$,μ,$\frac{\sigma}{\sqrt{n}}$,TRUE)

4 Sampling distribution of difference between sample means

Parameter:

$$\mu_1 - \mu_2$$

4.1 Assumptions

- The populations $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$
- σ_1 and σ_2 are known

4.2 The distribution of the difference between sample means

$$\bar{X}_1 - \bar{X}_2 \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$

Properties:

- $E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2$
- $Var(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$

4.3 Z-values

Since $\bar{X}_1 - \bar{X}_2$ is normally distributed, one can also use Z-values to calculate the sampling distribution.

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

4.4 Probabilities

Typical questions for this chapter requires the calculation of probabilities.

Steps:

1. $P(\bar{X}_1 - \bar{X}_2 < \bar{x}_1 - \bar{x}_2)$
2. Transform $\bar{x}_1 - \bar{x}_2$ into $Z \longrightarrow P(Z < z)$
3. Calculate probability by means of Z-table, HP calculator or Excel

HP calculator: $z \longrightarrow$ 

Excel: =NORM.S.DIST(z,TRUE) OR =NORM.DIST($\bar{x}$,μ,$\frac{\sigma}{\sqrt{n}}$,TRUE)

How do we know which value to use for $\bar{x}_1 - \bar{x}_2$?

What is the probability that the mean of group 1 is more/larger than the mean of group 2	$\bar{X}_1 > \bar{X}_2$	$\bar{X}_1 - \bar{X}_2 > 0$	use the value of 0
What is the probability that the mean of group 1 id less/smaller than the mean of group 2	$\bar{X}_1 < \bar{X}_2$	$\bar{X}_1 - \bar{X}_2 < 0$	use the value of 0
What is the probability that the mean of group 1 is equal to the mean of group 2	$\bar{X}_1 = \bar{X}_2$	$\bar{X}_1 - \bar{X}_2 = 0$	use the value of 0
What is the probability that the difference between the means of group 1 and group 2 is more/larger than c	$\bar{X}_1 > (\bar{X}_2 + c)$	$\bar{X}_1 - \bar{X}_2 > c$	use the value of c
What is the probability that the difference between the means of group 1 and group 2 us less/smaller than c	$\bar{X}_1 < (\bar{X}_2 + c)$	$\bar{X}_1 - \bar{X}_2 < c$	use the value of c

5 Questions

5.1 Question 1

The average time to complete a task is 727 milliseconds for Grade 3 students and 532 milliseconds for Grade 5 students. The variances for the groups are 12 000 and 10 000 respectively. These times are normally distributed. A random sample of 12 Grade 3 students and 14 Grade 5 students are selected.

1. What is the expected difference in mean times between Grade 3 and Grade 5 students?
2. What is the standard error of the difference in mean times between Grade 3 and Grade 5 students?
3. What is the probability that the mean time of Grade 3 students exceeds the mean time of Grade 5 students with 150 milliseconds or more?

5.2 Question 2

Why does the sampling distribution of the mean follow a normal distribution for a large enough sample size, even though the population may not be normally distributed?
Choose the correct answer:

(a) As the sample size becomes large enough for a distribution (that does not have any outliers), the sampling distribution of the mean is normally distributed.

(b) As the sample size gets large enough, the sampling distribution of the mean is approximately normally distributed.

(c) The sampling distribution of the mean is normally distributed for all sample sizes.

(d) As long as the distribution is approximately normally distributed, the sampling distribution of the mean is normally distributed for any sample size.

5.3 Question 3

An orange juice producer buys oranges from a large orange grove that has one variety of orange. The amount of juice squeezed from these oranges is approximately normally distributed, with a mean of 120 ml and a standard deviation of 7 ml. Given a sample of 36 oranges.

1. What is the probability that the sample mean amount of juice will be at least 128 ml?
2. The probability is 74% that the sample mean amount of juice will be contained between which two values symmetrically distributed around the population mean. Calculate these values.

5.4 Question 4

A study in Country A reported that 36% of companies have three or more female board directors. The following sample consists of 100 respondents.

1. What is the probability that the sample has between 31% and 40% of companies that have three or more female board directors?
2. Why can the normal distribution be used to approximate the distribution of this statistic?